If *hzn exchange status* fails, check that the hub containers are running with *docker ps* and that *HZN_EXCHANGE_URL* points to the correct IP address and port 3090.

Create your first Open Horizon service

An Open Horizon service is an ordinary Docker container plus a JSON service definition that tells the hub its name, version, architecture and image. This example builds *temp-sensor*, a small IoT service that reports a simulated temperature every 5 seconds.

Step 1 – Prepare signing keys and registry login: Services are signed so that agents can verify them.

```
hzn key create myorg student1@example.com      # creates
~/.hzn/keys/service.private.key and service.public.pem
export DOCKER_HUB_ID=<your-docker-hub-id>
docker login
mkdir ~/temp-sensor && cd ~/temp-sensor
```

Step 2 – Write the application: Save as *service.sh:*

```
#!/bin/sh
# Simulated IoT temperature sensor
while true; do
  TEMP=$(awk -v r=$RANDOM 'BEGIN{printf "%.1f", 20 + (r %
100) / 10}')
  echo "$(date -u +%FT%TZ) node=${HZN_DEVICE_ID}
temperature=${TEMP}C"
  sleep 5
done
```

The agent passes environment variables such as *HZN_DEVICE_ID* into every container, so the service can identify the device it runs on.

Step 3 – Write the Dockerfile:

```
FROM alpine:3.20
COPY service.sh /service.sh
RUN chmod +x /service.sh
CMD ["/service.sh"]
```

Step 4 – Build, test locally, and push the image:

```
docker build -t $DOCKER_HUB_ID/temp-sensor_amd64:1.0.0 .
docker run --rm -e HZN_DEVICE_ID=test $DOCKER_HUB_ID/temp-
sensor_amd64:1.0.0   # Ctrl+C to stop
docker push $DOCKER_HUB_ID/temp-sensor_amd64:1.0.0
```

For a Raspberry Pi, build a second image tagged *_arm64* with *docker buildx build --platform linux/arm64.*

Step 5 – Write the service definition: Save as *service.definition.json*, replacing *<your-docker-hub-id>:*

```
{
  "org": "myorg",
  "label": "temp-sensor for amd64",
  "url": "temp-sensor",
  "version": "1.0.0",
  "arch": "amd64",
  "public": true,
  "sharable": "singleton",
  "requiredServices": [],
  "userInput": [],
  "deployment": {
    "services": {
      "temp-sensor": {
        "image": "<your-docker-hub-id>/temp-sensor_
amd64:1.0.0",
        "privileged": false
      }
    }
  }
}
```

For larger projects, *hzn dev service* generates this folder structure, a Makefile and a horizon/directory of definitions automatically, and *hzn dev service starts* runs the service in a local simulated agent before you publish it.

Deploying the Open Horizon service

Deployment has three parts: publish the service to the Exchange, publish a deployment policy that says which nodes should run it, and give the node matching properties. The agbot then does the rest.

Publish the service: The CLI signs the definition with your key and stores it in the Exchange.

```
hzn exchange service publish -f service.definition.json
hzn exchange service list myorg/
```

Write the deployment policy: Save as *deployment.policy.json*. The constraint says 'deploy to any node whose sensor property equals temperature'.

```
{
  "label": "Temperature sensor deployment",
  "service": {
    "name": "temp-sensor",
    "org": "myorg",
    "arch": "*",
    "serviceVersions": [ { "version": "1.0.0" } ]
  },
```